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Las paginas restantes conservan el contenido util para revisar metodos, resultados, tablas y conclusiones.

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ock has improved the results of  
ed its use in everyday practice,  
d as a first-line treatment for this  
ible on ultrasound, which allows  
controlled obliteration of large veins under imaging control.

Foam sclerotherapy has been considered particularly attractive because it avoids the need for anesthesia, hospital admission, and long recovery times, is a low-cost procedure with faster recovery and minimal discomfort, and evidence in the literature shows its efficacy and safety. The procedure is performed under ultrasound guidance and involves the injection of a chemical agent (sodium tetradecyl sulfate or polidocanol) that acts more intensely on the intima because of its homogeneous distribution over the surface of the endothelium<sup>11</sup>. This action leads to the occlusion of the treated vein, removing its contribution to chronic venous hypertension, with subsequent healing of the ulcer and prevention of its recurrence<sup>12</sup>.

Although well documented for treating venous disease, published studies have primarily focused on the rates of occlusion of treated veins and the remission of CVI symptoms. Furthermore, most studies published in developed countries with greater public funding for health focus on the use of ablative therapies (laser and radiofrequency), to the detriment of the accessible and inexpensive method of foam sclerotherapy. Finally, no previous randomized study has compared foam sclerotherapy with or without elastocompression for VU healing. It is extremely plausible that combined sclerotherapy and elastocompression allows for higher rates of VU healing compared with elastocompression alone.

### Objectives

To evaluate the effect of sclerotherapy with polidocanol foam on UV healing in patients with CVI undergoing clinical treatment with elastocompression.

## METHODS

### Design

Single-center exploratory clinical trial with two parallel groups, 1:1 allocation, and blinding of the researchers responsible for measuring the main outcome. One group (Clinical group) will use only elastocompression (elastic bands), and the other (Foam group) will undergo treatment of CVI with polidocanol foam in the superficial venous system, associated with elastocompression. Elastocompression will be performed daily for 6 months in both groups.

### Study location

Vascular Surgery Outpatient Clinic of the Hospital de Clínicas de Itajubá (HCI), located in the city of Itajubá, Minas Gerais, Brazil. HCI is a quaternary hospital that is a reference in the macroregion of Southern Minas Gerais

### Inclusion criteria

CVI classified as CEAP C6 (active venous ulcer) by vascular surgeon;

Vascular mapping ultrasound with Doppler indicating chronic superficial venous insufficiency (varicose veins, reflux of saphenous and/or perforating veins);

Not having undergone previous surgical treatment of varicose veins;

Age between 18 and 80 years, of both sexes.

### Exclusion criteria

Polidocanol allergy;

Bronchial asthma;

Lack of agreement and signature of the free and informed consent form (TCLE);

Contraindication to the use of anticoagulants;

Coagulation disorder (INR > 1.5; platelets < 100,000)

Peripheral arterial occlusive disease (ankle brachial index < 0.8);

End-stage renal disease (CrCl < 15 mL /min);

Pregnancy;

Liver disease with cirrhosis;

History of recent deep vein thrombosis;

Immobility or confinement to bed;

Inability to receive outpatient care;

Heart failure - New York Heart Association classification II or III;

Infected ulcers or locoregional infectious cellulitis.

Lack of understanding of the therapeutic proposal;

Active neoplasia;

Diabetic foot (peripheral neuropathy or plantar ulceration).

Expected non-use of elastic band;

Documented or presumed thrombophilia ;

Nonrecanalised subacute or chronic deep vein thrombosis.

### Hypotheses

The null hypothesis ( $H_0$ ) is that there will be no differences in the healing rate between the Clinical and Foam groups at 6-month follow-up.

The alternative hypothesis ( $H_1$ ) is that the Foam group will have a higher healing rate at 6 months than the Clinical group (one-tailed).

### Sample size

We estimated that 93% of patients treated with foam sclerotherapy will have their VU healed, whereas only 68% of patients submitted to elastocompression alone<sup>13</sup>. The sample size for each group (Clinical and Foam) considering statistical proportions for the outcome "healing in 6 months", with a test power of 80%, assuming a significance level of 5%, is 30 patients for each group. Assuming a

|                              |   |   |   |   |   |
|------------------------------|---|---|---|---|---|
|                              | X |   |   |   |   |
|                              | X |   |   |   |   |
| Randomization and allocation | X | X | X |   |   |
| Follow-up                    | X | X | X | X |   |
| Data analysis                |   |   |   |   | X |
| Disclosure of results        |   |   |   |   | X |

recruitment sample of 34 is estimated for each group. The sample size was calculated using G\*Power software v. 3.1.9.4 (Universität Düsseldorf).

### Recruitment

Patients treated at the HCI vascular surgery outpatient clinic will be covered by the Unified Health System and will be referred from primary healthcare units. All patients who meet the inclusion criteria will be approached, and the reasons for the study will be explained during their routine or initial consultation. After agreeing to participate in the study, signing the informed consent form, and applying the exclusion criteria, the selected patients will undergo randomization.

### Randomization

Participants will be randomly selected for each of the two treatment groups through Randomizer (randomizer.org) to generate random numbers (1 or 2). Participants drawn in number 1 will be allocated to the Clinical group. Participants drawn in number 2 will be allocated to the Foam group. This process will be carried out until the 68th participant is reached.

### Allocation and blinding

Each patient will be allocated at a 1:1 ratio to their respective intervention groups and entered into a data acquisition spreadsheet.

There will be no blinding of the participants. The trained researchers performing the vascular Doppler ultrasound of the participants will be blinded. The data from the examination (performed following a specific study protocol) will be included in the CRF RedCap.

The researchers responsible for the statistical analysis will not contact the participants and will only have access to the data at the end of the intervention and monitoring stages.

### Elastocompression

Compression therapy is a fundamental part of CVI treatment, particularly in more severe cases involving edema, inflammation, or ulcers. Patients will be

13 cm wide × 2 m or 3 m long, depending on the size of the treated leg. Patients will be instructed to apply the bandage daily over the dressing. The bandage will be used in the morning and will last until the end of the day. Patients will be taught how to bandage the affected limb from the toe to the proximal third of the leg.

### Foam sclerotherapy

All sclerotherapy will be performed on an outpatient basis on a previously scheduled date. The patient will remain in the supine position with the limb elevated during the foam infusion. The puncture will be performed in the saphenous vein (great or small, according to the reflux mapping performed prior to the treatment) with a Jelco ® needle (number 18 G) or butterfly needle (number 21 G or 23 G), chosen according to the depth of the vein selected for puncture, guided by ultrasound. Local anesthesia using a "button" of 2% lidocaine without a vasoconstrictor will be used only to avoid pain during the puncture. The preferred puncture site will be where the vein is closest to the skin, in the distal third of the thigh or the proximal third of the leg. A mixture of 2 mL of 3% polidocanol and 8 mL of room air will be used, with foam produced using a three-way stopcock connected to two 10 mL screw-type syringes (Tessari technique) injected into the punctured vein, always with ultrasound monitoring. The volume of the mixture per session will be limited to 10 mL<sup>14</sup>.

### Vascular ultrasound

A vascular ultrasound examination with Doppler mapping of the veins of the affected limb will be performed, where the diameters of the superficial and deep veins, as well as their patency, compressibility, and flow direction will be recorded. The examinations will be performed by a vascular surgeon with at least 2 years of experience in vascular ultrasound.

### Criteria for discontinuing the study

The criteria for study participant discontinuation include severe bleeding, patient refusal, serious disease discovered (e.g. cancer), loss of follow-up, ulcer evolving

e will be the occurrence of VU healing after 6 months of follow-up.

#### *Secondary outcomes*

Ulcer healing time: period in days after treatment in which VU healing occurs.

Ulcer area reduction rate. The ulcer area will be calculated using a smartphone application before and during follow-up at 30 days, 3 months, and 6 months after treatment. The obtained result will be measured in cm<sup>2</sup>/month.

Adverse effects. The presence of hyperpigmentation, allergy, deep or superficial vein thrombosis, bleeding, or any other signs/symptoms reported by the participant will be assessed.

Pain. Assessed using a score of 0 to 10 on a visual analogue scale, after 30 days, 3 months and 6 months of intervention.

Assessment of quality of life. The results will be assessed using the EQ-5D© scale (EuroQol, The Netherlands, euroqol.org) before and during follow-up at 30 days, 3 months, and 6 months posttreatment.

Severity score assessment. It will be assessed using the Venous Clinical Severity Score (VCSS)<sup>15</sup> before and during follow-up, at 30 days, 3 months, and 6 months after treatment. The VCSS is scored from 0 to 24, where 0 indicates the least possible severity, and 24 indicates the worst possible.

#### *Data collection*

Participants will answer a protocol with structured questions and will be examined by HCI vascular surgeons. Clinical history and physical examination will be performed to assess the distribution of varicose veins and classification as well as attribution based on clinical signs, etiology, anatomy, and pathophysiology (CEAP)<sup>16</sup> of the disease and the Venous Disease Severity Score (VCSS). Data will be entered into a cloud-based spreadsheet (Google Sheet®) shared exclusively with the study researchers (Chart 1).

All selected patients will receive detailed information related to the use of elastic bands and ultrasound-guided polidocanol sclerotherapy.

#### *Monitoring*

There will be no data monitoring committee since the intervention will occur at a single point in the study, and the post-intervention follow-up will only be observational.

#### *Statistics*

Prism v.10 software (San Diego, CA, USA) will be used. The intention-to-treat principle will be considered.

for wound healing, patency, survival and limb salvage, we will use the Kaplan-Meier survival curve. The Log Rank test will be used to compare the curves. The significance level will be set at 5%.

#### *Ethical approval*

Ethical approval for this study was obtained from the Research Ethics Committee of the Itajubá School of Medicine (CAAE: 79577917.6.0000.5559, opinion number: 2,384,578). The universal trial registration number is U1111-1297-1789.

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**Individual contribution of the authors:**

Study conception and design: MAMS  
Data collection: HBGDO, MEMB  
Data analysis and interpretation: MAMS  
Manuscript writing: HBGDO, MEMB, MAMS  
Critical review of the text: Not applicable.  
Final approval of the manuscript\*: NLF, LAO, MAMS, RSF  
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Overall responsibility for the study: MAMS  
\*All authors read and approved the final version of the manuscript submitted for publication in HSJ.